KHALFAOUI AMINE

Cloud Computing & DevOps Engineering Student | Seeking a PFE Internship
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SUMMARY

Cloud Computing & Virtualization engineering student with hands-on experience in cloud infrastructure, CI/CD, monitoring and containerized applications. Strong interest in automation, scalability and reliable infrastructure. Looking for a PFE internship as a Cloud / DevOps Engineer.

TECHNICAL SKILLS

Cloud (AWS): EC2, ECR, ECS, Application Load Balancer, Auto Scaling Group, S3, Route 53, EBS, Security Groups
CI/CD & Code Quality: Jenkins, GitHub, Maven, Checkstyle, SonarQube (quality gate), Nexus Repository
Monitoring & Observability: Prometheus, Grafana, Loki, Promtail, Node Exporter, LangSmith
Kubernetes: Pods, Services, Ingress, Secrets, PersistentVolumeClaims, StorageClass, cluster networking
Containers & Networking: Docker (multi-container), Docker Compose, Docker networking, Nginx (API gateway / load balancer)
Backend Services: FastAPI, Apache Tomcat, Apache2, MySQL, PostgreSQL, MongoDB, Redis, Memcached, RabbitMQ, ARQ
Virtualization & IaC: Vagrant (Vagrantfile), VirtualBox, multi-VM environments, NFS shared storage
System Administration: Linux (Ubuntu), server networking, manual service configuration, remote management via SSH
Networking & Systems: Router and switch configuration, Active Directory (AD), firewall, network troubleshooting and supervision
Programming: Python, Java, JavaScript / Node.js, Next.js, PHP
Other: TensorFlow, Raspberry Pi, Bootstrap, AJAX

PROJECTS

CI/CD Pipeline with Jenkins, SonarQube & Docker on AWS (Nexus / Amazon ECR / ECS) — 2026
- Built a Jenkins CI/CD pipeline integrated with GitHub: code fetch, unit tests and Checkstyle with Maven, SonarQube analysis.
- Enforced a SonarQube quality gate, then automated the Docker image build and stored the artifact in a Nexus repository.
- Implemented a second variant pushing the image to Amazon ECR, with automated deployment to Amazon ECS.

Monitoring & Observability Stack on AWS (Prometheus, Grafana, Loki) — 2026
- Deployed and configured 4 Ubuntu EC2 instances: web server (Apache2), Prometheus, Grafana and Loki, each set up manually.
- Collected application metrics with Prometheus and centralized logs with Promtail pushing to Loki.
- Built Grafana dashboards on Prometheus and Loki data sources; restricted admin access to trusted IPs/VPN.

Kubernetes Microservices Deployment & AWS Persistent Storage — 2026
- Designed and deployed a multi-service app on Kubernetes: Apache Tomcat, RabbitMQ, Memcached and MySQL Pods.
- Configured Ingress, Services for inter-component communication, and Secrets for sensitive configuration.
- Provisioned persistent MySQL storage with a PVC and StorageClass on Amazon EBS.

Multi-Tier Application Infrastructure Automation with Vagrant — 2026
- Deployed a multi-tier infrastructure on 5 VirtualBox VMs: Nginx load balancers, Tomcat servers, RabbitMQ, MySQL, NFS.
- Built it manually first, then automated it with a Vagrantfile (IaC): reproducible, automated VM provisioning.

Multi-Tier DevOps Application Deployment on AWS — 2026
- Designed a scalable, cloud-native architecture using an Application Load Balancer and an Auto Scaling Group.
- Integrated backend services (MySQL, Memcached, RabbitMQ) with S3 storage and Route 53 DNS.

Microservices Deployment with Docker & Nginx (API Gateway) — 2026
- Deployed a distributed microservices application on 6 Docker containers (Java, Node.js, MySQL, MongoDB).
- Configured Nginx as an API gateway and load balancer; orchestrated the stack with Docker Compose and Docker networking.

INTERNSHIPS

AI_Powered_Jobs: Job Discovery & LLM Extraction Platform, Glorea Career — Summer 2026
- Built an asynchronous pipeline: a FastAPI API enqueues jobs in Redis, ARQ workers run discovery and LLM-based data extraction, and results are stored in PostgreSQL.
- Aggregated listings from multiple public job sources with deduplication and per-source status reporting.
- Configured worker concurrency, timeouts and limited retries, keeping the API responsive during background processing.

Network Administrator & Infrastructure Intern, Trésorerie Régionale des Finances de Tunis 3 — Summer 2026
- Documented the network infrastructure of the administration office.
- Configured routers and switches; managed users and resources with Active Directory.
- Managed firewall rules, troubleshot network issues and supervised the network.

Web Development Intern, MBC Training & Consulting — Summer 2025

- Built a web platform for searching job, internship and other offers with PHP & MySQL, including authentication.
- Developed a responsive user interface using Bootstrap, JavaScript and AJAX.

Embedded Computer Vision System, PFE — 2024

- Built an embedded computer vision system with Python and TensorFlow on Raspberry Pi for real-time image prediction.
- Enabled remote device management, monitoring and control over SSH.

EDUCATION

Engineering Degree in Cloud Computing & Virtualization — Iteam University, 2024 – 2027
Bachelor in Software Engineering — ISSAT Kasserine, 2021 – 2024

LANGUAGES

English: B1 | **French:** A2

COMMUNITY COMMITMENT

- Treasurer — IEEE Student Branch